CASCADIA ANALYTICS

Amprius Technologies

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Contract-Manufacturing Transition Accelerates Revenue but Current Multiples Already Discount Flawless Execution

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Amprius Technologies has successfully engineered a commercial inflection across its silicon-anode cell portfolio, lifting gross margins to 27% in Q2 FY2026 while expanding top-line guidance to at least $140M on the back of major wins in unmanned aerial systems and light electric vehicles. However, the market has detached from underlying cash conversion realities, overlooking an acute working capital drain evidenced by first-half operating cash flow of -$40.1M and substantial structural friction from legacy plant cancellations. At an equity valuation trading at 18.2x latest fiscal year revenue, the risk-reward profile is skewed decisively downward. Our probability-weighted discounted cash flow model establishes a blended target price of $5.84, representing -36.1% upside versus the reference price of $9.14.

HOLD	$5.84	-36.1%	$9.14
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

While consensus remains fixated on top-line percentage growth and high-profile contract announcements, public markets are ignoring the severe cash flow drag generated by surging accounts receivable and contract-manufacturing working

capital requirements. Long-term value creation depends entirely on normalizing free cash flow conversion rather than top-line run-rates alone.

Executive summary

Amprius operates as a specialized silicon-anode lithium-ion battery developer targeting drones, aviation, and electric vehicles with high energy density cells. Over recent quarters, the organization has pivoted away from capital-intensive direct factory builds toward an asset-light contract-manufacturing structure centered on its SiCore platform. This strategic reallocation has allowed the company to rapidly fulfill surging purchase commitments across commercial and government aerospace channels without absorbing massive equipment lead times. Commercial validation has followed, evidenced by expanding relationships with tier-one unmanned aerial system (UAS) manufacturers and initial commercial entries into light electric vehicles (LEVs).

The operational data through the first half of FY2026 highlights an accelerating top line coupled with fundamental unit-level expansion. In Q2 FY2026, revenue expanded 126% year-over-year to $34.0M, following a 153% surge to $28.5M in Q1 FY2026 and a 137% rise to $25.2M in Q4 FY2025. Gross margins have recovered from negative 21% in the year-ago period to 27% in Q2 FY2026, enabling management to increase full-year FY2026 revenue guidance to at least $140M and gross margin guidance to at least 28%. Key commercial milestones include a multi-year supply agreement exceeding $100M with Stark Future, a $24M SiCore purchase order from a European drone developer, and defense contract expansion with the Defense Innovation Unit (DIU) to $18.1M. Adjusted EBITDA has shown sharp improvements, registering at -$1.0M in Q2 FY2026 as the business approaches corporate breakeven.

Despite these commercial advancements, underlying cash flow dynamics remain constrained by aggressive working capital demands and legacy restructuring liabilities. Amprius reported first-half FY2026 operating cash flow of -$40.1M, with accounts receivable expanding by $16.8M over the same period. In Q1 FY2026 alone, cash burn totaled $37.3M, burdened by a $20.6M Colorado lease settlement payment and an $11.5M expansion in trade receivables. This builds upon the significant restructuring charges incurred at the end of FY2025, which included a $22.5M asset impairment alongside a $20M lease exit commitment associated with the cancelled Colorado facility. The balance sheet closed FY2025 with $90.5M in cash against $39.9M in total debt, leaving net cash of $50.6M. Continued cash conversion friction requires a balanced stance, supporting our Hold rating.

Operational snapshot

$9.14	$71.9M	$-35.5M	-49.4%	1.80	$9.14
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Proprietary and Contractual

The competitive moat protecting Amprius rests on the technical performance of its high energy density silicon-anode cells, which solve flight endurance constraints that conventional graphite chemistry cannot address. In defense and high-altitude aviation environments, battery weight directly dictates payload capacity, sensor suite capabilities, and operational range. By scaling its SiCore platform, Amprius has established switching costs that prevent aerospace customers from substituting lower-density alternatives without requiring full airframe redesigns. In Q3 FY2025, aviation customers represented 75% of total quarterly revenue, underscoring how deeply embedded these cells are inside high-altitude and unmanned aerial systems.

Government procurement standards and compliance mandates form a second durable barrier to entry. Amprius demonstrated compliance with 11-of-11 National Defense Authorization Act (NDAA) internal component standards in Q4 FY2025, creating an insulated competitive position across United States defense programs. The expansion of its Defense Innovation Unit contract from $14.8M to $18.1M illustrates growing federal reliance on domestic cell architectures. Furthermore, United States defense customers utilizing Amprius components secured roughly $500M in new military awards during Q1 FY2026, generating long-cycle program visibility that insulated the core defense business from commercial pricing pressure.

Commercial relationships in adjacent sectors have added multi-year volume commitments that reinforce supply chain stickiness. The multi-year supply agreement with Stark Future, valued at over $100M, and the $24M contract for SiCore cells with a European drone developer confirm that technical qualification cycles yield multi-year revenue tails. Because qualifying high energy density cells into complex electric systems demands extensive thermal, safety, and reliability benchmarking, customer churn remains exceptionally low once high-volume delivery commences.

Nevertheless, the durability of this moat faces tests as contract-manufacturing scale introduces potential manufacturing margin friction. While gross margins expanded to 27% in Q2 FY2026, the first quarter of FY2026 experienced gross margin compression to 20% compared to 24% in Q4 FY2025, driven by product mix shifts toward higher-volume light electric vehicle programs. Maintaining pricing authority while licensing or contract-manufacturing outside proprietary facilities remains an ongoing execution risk that dictates whether economic returns persist long-term.

Valuation

Three scenarios make the valuation range explicit

To evaluate the equity intrinsic value of Amprius Technologies, Cascadia Analytics utilizes a three-scenario discounted cash flow framework covering the forecast window from FY2026

through FY2030. Our baseline scenario assumes a compound annual revenue growth rate of 38.5% through FY2030, anchored by execution against the current $140M FY2026 top-line guidance and multi-year production commitments. In this base case, operating leverage across administrative overhead allows target free cash flow margins to reach 14.3% in the terminal forecast period, yielding a fair equity value of $5.26 per share, representing a -42.5% contraction relative to the reference spot price of $9.14.

In our downside bear scenario, commercial execution is compromised by customer vertical concentration and extended working capital absorption. This case models an 18.0% revenue CAGR with final-year free cash flow margins stabilizing at 9.3%, producing an equity value of $2.12 per share, or -76.8% versus spot. Conversely, our bull scenario models uninhibited contract-manufacturing scale across aviation, defense, and electric vehicles, generating a 55.0% revenue CAGR and expanding target free cash flow margins to 19.5%. The bull scenario implies a target price of $10.73 per share (+17.4% upside). Weighting these outcomes 25% bear, 50% base, and 25% bull produces a probability-weighted blended target price of $5.84, which sits -36.1% below current trading levels.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	18.0%	9.3%	$2.12	-76.8%
Base	50%	38.5%	14.3%	$5.26	-42.5%
Bull	25%	55.0%	19.5%	$10.73	+17.4%

Bear $2.12

Aviation and defense contract lumpy ordering patterns materialize alongside extended trade credit cycles, suppressing cash generation. Elevated capital requirements and pricing concessions on light electric vehicle agreements restrict terminal free cash flow margins to 9.3% on an 18.0% revenue CAGR.

Base $5.26

Contract manufacturing delivers predictable volume ramp across drone and light electric vehicle accounts without necessitating direct factory capital expenditure. Top-line expands at a 38.5% CAGR through FY2030 while free cash flow margin normalizes at 14.3%.

Bull $10.73

Commercialization across global electric vehicle and high-altitude defense programs accelerates, yielding a 55.0% revenue CAGR. Manufacturing efficiencies and operating cost control push terminal free cash flow margins to 19.5%, justifying multiple expansion.

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Valuation framework

A live discount rate, not a static hurdle

We discount projected cash flows at a weighted average cost of capital of 11.18%, reflecting an elevated risk profile that accounts for early commercialization execution, customer concentration, and historical cash burn. The equity cost of capital incorporates an equity beta input of 1.80, capturing significant equity volatility, cyclicality in aviation capital budgets, and market sensitivity to working capital absorption. Over the forecast horizon through FY2030, this beta is modeled to decay gradually toward mature hardware supplier benchmarks as cash flow profitability solidifies.

The terminal value calculation applies a 3.0% perpetual growth rate in our base case scenario, consistent with long-run global economic expansion and steady baseline demand across specialized aerospace components. This hurdle rate appropriately discounts future free cash flow contributions back to present value across 143.5 million diluted shares, avoiding the aggressive cost of capital compression often applied to high-growth battery equities.

FCF margin normalization is the engine

The transition from severe historical cash burn toward normalized free cash flow conversion represents the central driver within our quantitative model. For FY2025, the company recorded negative free cash flow of -$35.5M on revenue of $71.9M, representing a negative free cash flow margin of -49.4%, driven by negative operating margins of -32.0% and inventory builds. The strategic decision to exit the Colorado production facility, despite incurring a $22.5M non-cash asset impairment and a $20.6M cash lease exit settlement in early FY2026, structurally insulates future periods from major internal manufacturing CapEx burdens.

Our valuation framework explicitly maps the inflection of unit economics into normalized cash flows. With gross margins demonstrating resilience at 27% in Q2 FY2026 and targeted to exceed 28% for full-year FY2026, operating leverage begins to offset fixed corporate research and administrative spending. Across our forecast horizon, target free cash flow

margins reach 9.3% in the bear case, 14.3% in the base case, and 19.5% in the bull case. Achieving these margins requires working capital stabilization, particularly reversing the first-half FY2026 trend where accounts receivable consumed $16.8M.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2030**
Live WACC	**11.18% (2026-10-29)**
Terminal growth	**3.0%**
Equity beta	**1.80**
Diluted shares	**143.5 million**

HOLD

over $5.84 blended target

Competitive moat and concentration risk

We rate the moat as None and we view the market structure as Fragmented (batteries). We assign a discount of 10% to sector median multiples on moat strength alone.

- demonstrated energy-density lead, ~450 Wh/kg shipped cells vs ~260 Wh/kg for conventional cells, with Q3 2025 revenue +173% YoY to $21.4m (2025-2026 coverage) .
- The inflection is real but thin, FY2025 revenue $73.0m (+202%) with gross margin turning to +11% for the year and 24% in Q4 2025, and positive adjusted EBITDA in Q4 2025 (Wedbush/company coverage, Mar 2026). One quarter of positive gross margin after years of negative margins is proof of unit economics, not pricing power.
- The weakest element is everything else about durability, contract-manufacturing dependence (Korea Battery Alliance, 2+ GWh, only 30-40% utilised for 2026 guidance), one customer at $27.1m of 2025 revenue, and mass-market EV economics unproven (EV buyers optimise $/kWh, not Wh/kg). A technology lead that has not yet survived scale-up is not yet a moat.

On a multiple cross-check, the justified range is 9.8-11.5x / 17.0-20.0x. The cross-check value is $1.70-$2.00 vs Base $6.87, spot $8.56, on a base of Framework base from the October 27 moat report. The gap verdict is Not at all, market already pays far more than moat justifies (spot P/E 85.6x).

Backlog is $53.3m RPO, Q3 2025, stale, with coverage of 4.6 months vs 2026 guide floor and visibility rated Low. On backlog alone, the stance is Neutral, model relies on future wins, not booked backlog.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	Excluded
counterparty credit and funding dependence	4 points
single product, single program, or single platform dependency	8 points
single supplier, single fab, or single geography exposure	Excluded
platform, channel, or distribution dependency	1 points
customer insourcing risk	1 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 22 percent. Shown side by side with the moat premium or discount, the net stance is -30 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The following families are excluded under the double-count rule because they rest on the same fact already reflected in the moat discount, and they contribute no points here. customer concentration, single supplier, single fab, or single geography exposure.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on SiCore Platform Commercialization and Defense Aviation Deployments HOLD

The commercial trajectory of Amprius centers on deploying its high energy density silicon-anode cells into specialized mobility applications where alternative chemistries fall short. Commercial revenue generation has consolidated around two core operational pillars, consisting of unmanned aerial systems and aviation platforms alongside emerging high-volume light

electric vehicle contracts. The SiCore platform has served as the key technical vehicle facilitating this commercial inflection, demonstrated by second-generation SiCore cell shipments expanding fourfold year-over-year in Q3 FY2025.

Aviation and defense integration provides immediate baseline volume and margin stability. In Q3 FY2025, 75% of quarterly revenue originated from aviation, encompassing unmanned aerial vehicles and high-altitude platforms. This vertical concentration was punctuated by a $35M follow-on purchase order from a leading UAS manufacturer in Q3 FY2025, followed by expansion in the DIU contract value to $18.1M in Q1 FY2026. The full adherence to 11-of-11 NDAA-compliant internal component requirements provides a protected status across Department of Defense procurement budgets, supporting defense clients that recently captured $500M in new military contracts.

Simultaneously, light electric vehicle diversification offers significant top-line expansion, albeit at variable margin profiles. The multi-year supply agreement with Stark Future exceeding $100M demonstrates that high energy density silicon-anode technology can migrate into specialized ground mobility. However, product mix shifts toward high-volume commercial contracts require careful oversight. Gross margins experienced a temporary step-back from 24% in Q4 FY2025 to 20% in Q1 FY2026 as product delivery schedules shifted toward high-volume applications before recovering to 27% in Q2 FY2026.

Customer expansion has demonstrated measurable momentum across recent quarters. During Q3 FY2025, Amprius delivered shipments to 159 end customers, including 80 newly acquired accounts, illustrating broadening commercial validation. Maintaining this trajectory while converting trial volumes into contracted high-volume supply remains vital for absorbing corporate operating expenses, which have thus far demonstrated disciplined containment relative to expanding sales volumes.

What to monitor

- **Accounts receivable and working capital consumption** Operating cash flow registered a negative $40.1M during the first half of FY2026, accompanied by an accounts receivable increase of $16.8M, representing a critical operational liquidity bottleneck that requires normalization.
- **Contract manufacturing gross margin stability** Gross margin progression from 20% in Q1 FY2026 to 27% in Q2 FY2026 against full-year targets of at least 28% must be maintained amid shifting commercial delivery mixes between defense aviation and commercial mobility accounts.
- **Full-year adjusted EBITDA inflection** Execution against management targets calling for at least $4M in full-year FY2026 positive adjusted EBITDA following the -$1.0M result in Q2 FY2026 and -$1.8M in Q1 FY2026.
- **Contract conversion across military customer pipelines** Conversion of the $500M defense customer order pipeline into incremental follow-on orders for Amprius cells beyond the current $18.1M DIU allocation.

+205.1%	11.5%	-32.0%	-49.4%	$50.6M	-42.4%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$71.9M
Gross Profit	$8.3M
EBIT	$-23.0M
Net Income	$-44.0M
EBITDA	$-18.7M
Free Cash Flow	$-35.5M
Diluted EPS	$-0.35
ROE	-42.4%

Risk & moat matrix

The bear case

SHARE EROSION

Failure to maintain delivery timelines under large contract-manufacturing arrangements could prompt UAS and aerospace customers to explore alternate cell suppliers. Given that a single vertical represented

M&A DIGESTION

Executing rapid operational realignments following the termination of the Colorado production facility creates supply chain coordination risks. Operational execution must proceed seamlessly through

REGULATORY AND PRICING SCRUTINY

Shifting production volumes toward commercial light electric vehicles exposes unit pricing to higher volume sensitivity. As seen in Q1 FY2026 when gross margin stepped back to 20%, product mix

TECHNOLOGY DISRUPTION

Rapid advances in competing solid-state or alternative silicon battery architectures could reduce Amprius's technological performance advantage. Continued leadership depends on sustaining density advantages

REINVESTMENT TREADMILL

Unplanned working capital requirements create ongoing liquidity drains despite achieving accounting gross profitability. First-half FY2026 operating cash burn of -$40.1M indicates that contract fulfillment demands substantial trade credit support.

WACC AND DURATION SENSITIVITY

With an equity beta of 1.80 and a WACC of 11.18%, valuation targets remain highly vulnerable to capital market volatility. Given that the company's valuation multiple stands at 18.2x latest fiscal year

75% of Q3 FY2025 revenue, any supplier substitution would immediately impair corporate run-rates.

external contract manufacturers to prevent delivery interruptions.

shifts toward commercial mobility can dilute blended margin realization.

across second-generation SiCore architectures.

sales, any discount rate re-rating impairs equity multiples sharply.

The strengths

OLIGOPOLY POSITION

Amprius possesses proprietary high energy density silicon-anode cells qualified into flight-critical defense and aerospace platforms. These cell architectures cannot be readily displaced without clients undergoing costly platform engineering and recertification cycles.

FCF COMPOUNDING

The asset-light contract-manufacturing model eliminates heavy internal factory construction CapEx requirements going forward. With operating expenses decoupled from revenue growth, gross margins exceeding 28% establish a framework for long-term cash flow leverage.

GEOGRAPHIC DURABILITY

Full adherence to 11-of-11 NDAA-compliant internal component requirements insulates Amprius within domestic defense procurement allocations. Continued alignment with United States Department of Defense initiatives provides durable demand independent of commercial economic cycles.

PIPELINE OPTIONALITY

The contract-manufacturing footprint supports rapid expansion across diverse mobility verticals without direct capital expenditure. Recent contract awards including the $100M+ Stark Future agreement illustrate scalability across both aerospace and ground transportation applications.

The bottom line on risk and reward

Amprius Technologies has successfully engineered a vital operational turnaround, pivoting to a capital-light contract-manufacturing strategy that has ignited top-line expansion, expanded gross margins to 27% in Q2 FY2026, and narrowed adjusted EBITDA losses to -$1.0M. Management guidance indicating full-year FY2026 revenue of at least $140M and positive adjusted EBITDA of at least $4M demonstrates that the commercialization of high energy density silicon-anode cells has moved past early proof-of-concept stages into multi-year commercial adoption.

However, investment analysis must separate operational progress from equity valuation discipline. Trading at 18.2x latest fiscal year revenue with an equity market capitalization of

$1.31B, market pricing currently discounts flawless long-term growth execution while ignoring immediate balance sheet friction. Cash outflows remain elevated, highlighted by -$40.1M in first-half FY2026 operating cash flow, driven by surging accounts receivable balances and Colorado lease exit disbursements totaling $20.6M. With cash reserves at $90.5M at the end of FY2025 and ongoing operational working capital needs, cash conversion remains the paramount risk.

Our quantitative discounted cash flow framework, which models an 11.18% cost of capital and evaluates cash generation through FY2030, yields a base case equity target of $5.26 and a probability-weighted blended target of $5.84. This blended target price indicates -36.1% downside relative to the reference spot price of $9.14. Consequently, we maintain a Hold (Valuation Concerns) rating, advising investors to await a more advantageous valuation entry point or tangible proof of sustainable free cash flow generation before committing capital.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

SiCore High Energy Density Platform: Capital-Light Silicon-Anode Deployment

The technological centerpiece of Amprius's commercial expansion is its SiCore cell family, designed to deliver industry-leading energy density for aerospace, unmanned aerial systems, and high-performance mobility applications. Historically, silicon-anode cell developers struggled with structural expansion during lithiation cycles, preventing commercial mass production. Amprius addressed this technical challenge by scaling its proprietary chemistry through validated third-party contract manufacturers, eliminating internal facility build-out risks while delivering commercial cells that satisfy demanding flight-duration criteria.

The decision executed in late FY2025 to cancel the planned Colorado production facility, resulting in a $22.5M non-cash asset impairment and a $20.6M lease settlement payment executed in Q1 FY2026, cemented the operational pivot toward third-party manufacturing partnerships. This restructuring cleared the path for unit-level margin improvements. Gross margins transitioned from negative 21% in Q4 FY2024 to positive 24% in Q4 FY2025, reaching 27% in Q2 FY2026 as production volumes scaled across manufacturing partners.

Commercial deployment of SiCore cells has expanded across both defense aviation and high-performance commercial ground mobility. In the aviation domain, the SiCore platform secured a $35M follow-on purchase order from a major UAS manufacturer in Q3 FY2025, alongside a $24M order from a European drone developer in Q2 FY2026. Simultaneously, the signing of a multi-year supply agreement exceeding $100M with Stark Future confirms the viability of SiCore cells in demanding light electric vehicle applications. With second-generation SiCore cell volume shipments expanding fourfold year-over-year in Q3 FY2025, the platform represents the foundational core of Amprius's commercial growth.

What the recent record shows

- **Second-generation SiCore volume scale** Shipments of second-generation SiCore cells expanded fourfold year-over-year in Q3 FY2025, validating contract-manufacturing execution.
- **Major European commercial drone commitment** Secured a $24M purchase order for SiCore cells from an established European drone developer in Q2 FY2026.
- **Light electric vehicle expansion** Executed a multi-year supply contract valued at more than $100M with Stark Future to deliver high energy density battery cells.
- **NDAA compliance verification** Confirmed complete adherence across 11-of-11 NDAA-compliant internal component requirements in Q4 FY2025, reinforcing sole-source defense positioning.

> **Our read:** The SiCore platform establishes Amprius as a technically superior cell provider within high-performance defense aviation and commercial niches. However, because current equity multiples fully discount this success without accounting for working capital absorption, capital preservation demands a Hold posture until operating cash flows turn sustainably positive.